Alexander (Sasha) Apartsin, Ph.D.

Senior Lecturer, School of Computer Science, Faculty of Sciences · Holon Institute of Technology (HIT)

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Machine-learning-systems researcher with two decades of building and measuring AI at production scale, now working at the intersection of **energy-efficient AI computation**, measurement methodology, and optimization. Principal investigator for the JouleFlex proposal on energy-optimal power control for AI data centers, committing about one day per week across the 24-month project.

Fit to an energy-and-AI research program. Four threads of this record converge on the proposed work: (1) **production-scale performance-versus-cost engineering**: CPU-load models for cost-efficient system sizing at Amdocs that contributed to about $10M in savings, the closest analogue to the project's techno-economic thread; (2) delivery of **competitively funded, milestone-based R&D** as the responsible partner on a BIRD Israel-US grant and an Israel Innovation Authority Magneton grant; (3) a peer-reviewed lineage in **energy-efficient measurement under noise** (IEEE JSTARS time-of-flight), mapping onto the three-tier NVML/RAPL/BMC instrumentation rigor; (4) **energy-efficient inference** (selective inference in CalexNet) and a US inference-optimization patent. The PI built and validated the measurement study the project extends.

Research expertise

- Energy-efficient AI inference
- ML systems & measurement
- Benchmarking methodology
- Optimization
- Signal processing under noise
- Computer vision
- LLMs & generative AI
- Anomaly detection
- Multi-agent coordination

Bibliometrics

9	500+	3	7
H-INDEX (SCHOLAR)	CITATIONS	GRANTED US PATENTS	PATENT APPLICATIONS

Education

Ph.D., Computer Science	Tel-Aviv University · Machine Learning & Signal Processing	2012
Post-doctoral Fellow	Weizmann Institute of Science · Computer Vision (host Dr. Anat Levin)	2012–13

M.Sc., Management (Information Systems)	Polytechnic Institute of NYU, Israel Extension	2006
M.Sc., Computer Science	Weizmann Institute of Science · Computer Vision (adv. Prof. Ronen Basri)	2005
B.Sc., Computer Science	Technion, Israel Institute of Technology	1995

Professional certifications: IEEE CSDP (2008); PMI PMP (2009).

Academic appointments

Senior Lecturer (equivalent)	Holon Institute of Technology (HIT), Computer Science	2024–now
Head, Software Engineering Track	HIT, School of Computer Science	2024–25
Adjunct Lecturer	Bar-Ilan (2022), Tel-Aviv University (2013, 2015), Academic College of Tel-Aviv-Yaffo (2011)	2011–22

Selected industry experience

Head of Data Science	Questar Automotive (SafeRide) · AI for predictive maintenance at fleet scale	2020–24
Vice President, Artificial Intelligence	Harmon.IE · AI for enterprise document management	2019–20
Head of AI Research; Distinguished Member of Technical Staff	Motorola Solutions Israel Innovation Center	2015–19
AVP, Data Science	Citigroup TLV Innovation Lab · ML for trading-flow analytics	2013–15
Data Scientist	Amdocs · capacity modeling for high-scale distributed systems	2010–11

Earlier: CTO/Founder SearchNG (distributed search); algorithm and architecture roles at Comverse, McGraw-Hill (Tegrity), Product Computers, SmartLight.

Selected publications

Grant-relevant work first; full list of 20+ peer-reviewed articles at Google Scholar.

Energy-efficient time-of-flight estimation in the presence of outliers: a machine-learning approach. Apartsin, Cooper & Intrator. IEEE J. Selected Topics in Applied Earth Observations & Remote Sensing, 7(4), 2014. WoS Q1

A benchmark for evaluating diagnostic questioning efficiency of LLMs in patient conversation. Werthaim, Kimhi, Apartsin & Aperstein. Scientific Reports, 2026. WoS Q1

IRC-Bench: recognizing entities from contextual cues. Aperstein, Moran & Apartsin. Machine Learning & Knowledge Extraction, 8(7), 2026. WoS Q1

CoEval: ranking language models for custom tasks without labeled data or trustworthy benchmarks. Apartsin & Aperstein, 2026 (under review).

CalexNet: soft cascade-aligned training and calibration for lightweight early-exit branches. Aperstein & Apartsin. Electronics, 15, 2026. **WoS Q2**

Mapping license-plate recoverability under extreme viewing angles for opportunistic urban sensing. Adamenko, Ben-Aharon, Aperstein & Apartsin. AI (MDPI), 7(7), 2026. **WoS Q1**

Time-of-flight estimation in the presence of outliers, Parts I & II. Apartsin, Cooper & Intrator. IEEE Trans. Geoscience & Remote Sensing, 52(6–7), 2013. **WoS Q1**

Accurate blur models vs. image priors in single-image super-resolution. Efrat, Glasner, Apartsin, Nadler & Levin. IEEE ICCV, 2013. (230+ citations)

Book: *Hands-On AI Science*, Apartsin & Aperstein.

Patents (selected)

- **System and method for the unification and optimization of machine-learning inference pipelines** — US 2024/0013905 A1 (application). Directly relevant to inference-systems efficiency.
- System and method for physical model-based machine learning — US 2022/0382939 A1 (application).
- System and method for model-configuration selection — US 11,868,899 B2 (granted, 2024).
- Response determination with public-safety impact — US 10,825,450 B2 (granted, 2020).
- Matching portions of input images — US 8,175,412 B2 (granted, 2012).

Grants & awards

- **BIRD** (Israel–US Binational Industrial R&D Foundation), group leader, 2018 — on-device video analytics.
- **Israel Innovation Authority — Magneton**, group leader, 2017 — academia–industry transfer (with Tel-Aviv University).
- Citi Lablink Venture Grant, leader, 2015; Israel Innovation Authority incubation grant, founder, 1999.
- Informa Tech "Automotive AI Product of the Year", 2021 (Questar / SafeRide).
- Don & Sara Marejn Foundation award for outstanding Ph.D. achievement, Tel-Aviv University, 2011.

Teaching & mentoring

Graduate and undergraduate courses at HIT: Large Language Models in Action, Practical Generative AI Models for Visual and Signal Data, Image Processing and Computer Vision with Deep Learning, NLP for Healthcare, Introduction to Computer Science; the *Hands-On AI Science* course series. Established two Industry Advisory Boards defining graduate-readiness competencies; supervises a large cohort of applied AI student projects.

Current research direction

Energy-optimal operation of AI accelerators: measuring where energy per unit of useful compute is minimized on GPUs (the "Joule Point"), modeling the underlying power–time tradeoff, and turning it into fleet-design, control, and grid-facing decisions. The foundational measurement study is a manuscript under review; the JouleFlex proposal to the Israeli Ministry of Energy extends it into an open dataset, prediction models, and a decision-support toolkit for energy-constrained AI data centers.

Prepared for the Israeli Ministry of Energy competitive R&D call (34/2026). Full official CV available on request. Bibliometrics from Google Scholar; publication tiers from Web of Science / SCImago.